Exercise JA

Paragraph

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$. A straight line L, perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$

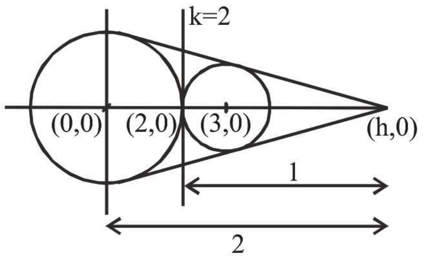
\n

Q1. A possible equation of L is :

(A) $x - \sqrt{3}y = 1$ (B) $x + \sqrt{3}y = 1$ (C) $x - \sqrt{3}y = -1$ (D) $x + \sqrt{3}y = 5$

Ans. A

Sol. $h = \dfrac{2\times 3 - 1\times 0}{2 - 1} = 6$



equation of tangents from (6, 0) :

$y - 0 = m(x - 6) \Rightarrow y - mx + 6m = 0$

use $p = r$

$$\left|\frac{6m}{\sqrt{1+m^2}}\right| = 2 \Rightarrow 36m^2 = 4 + 4m^2$$

$$32m^2 = 4$$

$$m^2 = \frac{1}{8}$$

$$\Rightarrow m \pm \frac{1}{2\sqrt{2}}$$

at $m = \dfrac{1}{2\sqrt{2}}$

equation of tangent will be $x + 2\sqrt{2}y = 6$

Paragraph

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$. A straight line L, perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$

\n

Q2. A common tangent of the two circles is :

(A) $x = 4$ (B) $y = 2$ (C) $x + \sqrt{3}y = 4$ (D) $x + 2\sqrt{2}y = 6$

Ans. D

Sol. Equation of tangent at P will be $\sqrt{3}x + y = 4$

Slope of line L will be $\frac{1}{\sqrt{3}}$

Let equation of L be : $y = \frac{x}{\sqrt{3}} + c$

$\Rightarrow x - \sqrt{3}y + \sqrt{3}c = 0$

Now this L is tangent to 2^{nd} circle

So $\frac{3+\sqrt{3}c}{2} = \pm \quad \Rightarrow \quad c = -\frac{1}{\sqrt{3}}$

or $c = -\frac{5}{\sqrt{3}}$

using $c = -\frac{1}{\sqrt{3}}$

$y = \frac{x}{\sqrt{3}} - \frac{1}{\sqrt{3}} \quad \Rightarrow \quad x - \sqrt{3}y = 1$ Hence (A)

Q3. The locus of the mid–point of the chord of contact of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ is :

(A) $20(x^2 + y^2) - 36x + 45y = 0$
(B) $20(x^2 + y^2) + 36x - 45y = 0$
(C) $36(x^2 + y^2) - 20x + 45y = 0$
(D) $36(x^2 + y^2) + 20x - 45y = 0$

Ans. A

Sol. Let mid point be (h, k),
then chord of contact :
$hx + ky = h^2 + k^2$ ....(i)
Let any point on the line $4x - 5y = 20$ be $\left(x_1, \frac{4x_1 - 20}{5}\right)$

$\therefore$ Chord of contact :

$5x_1x + (4x_1 - 20)y = 45$ ....(ii)

(i) and (ii) are same

$\therefore \frac{5x_1}{h} = \frac{4x_1 - 20}{k} = \frac{45}{h_2 + k_2}$

$\Rightarrow x_1 = \frac{9h}{h^2 + k^2}$ and $x_1 = \frac{45k + 20(h^2 + k^2)}{4(h^2 + k^2)}$

$\Rightarrow \frac{9h}{h^2 + k^2} = \frac{45k + 20(h^2 + k^2)}{4(h^2 + k^2)}$

$\Rightarrow 20(h^2 + k^2) - 36h + 45k = 0$

$\therefore$ Locus is $20(x^2 + y^2) - 36x + 45y = 0$

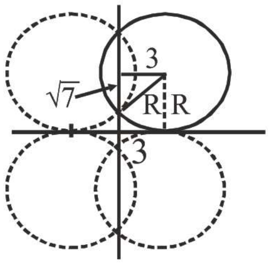
Q4. Circle(s) touching x–axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ on y–axis is (are) :

(A) $x^2 + y^2 - 6x + 8y + 9 = 0$ (B) $x^2 + y^2 - 6x + 7y + 9 = 0$

(C) $x^2 + y^2 - 6x - 8y + 9 = 0$ (D) $x^2 + y^2 - 6x - 7y + 9 = 0$

Ans. A, C

Sol.



As per figure,

$R^2 = 3^2 + (\sqrt{7})^2$

$\Rightarrow R = 4$

$\therefore$ centre $\equiv (3, 4)$

radis 4

$\therefore$ equation $x^2 + y^2 - 6x - 8y + 9 = 0$

such a circle can lie in all 4 quadrants as shown in figure.

$\therefore$ equation can be $x^2 + y^2 \pm 6x \pm 8y + 9 = 0$

Q5. A circle S passes through the point (0,1) and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then

(A) radius of S is 8 (B) radius of S is 7 (C) centre of S is (–7,1)

(D) centre of S is (–8,1)

Ans. B, C

Sol. Let the cirlce be

$x^2 + y^2 + 2gx + 2fy + c = 0$ ...(1)

given circles

$x^2 + y^2 - 2x - 15 = 0$ ...(2)

$x^2 + y^2 - 1 = 0$ ....(3)

(1) & (2) are orthogonal

$\Rightarrow -g + 0 = \frac{c - 15}{2}$

$\Rightarrow c = 1 \,\&\, g = 7$

so the cirle is

$x^2 + y^2 + 14x + 2fy + 1 = 0$ it passes thrgouh

$(0, 1) \Rightarrow 0 + 1 + 0 + 2f + 1 = 0$

$$f = -1$$

$$\Rightarrow \quad x^2 + y^2 + 14x - 2y + 1 = 0$$

Centre $(-7, 1)$

radius = 7

Q6. Let RS be the diameter of the circle $x^2 + y^2 = 1$, where S is the point (1, 0). Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q. The noraml to the circle at P intersects a line drawn through Q parallel to RS at point E. Then the locus of E passes through the point(s)

(A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$ (B) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (C) $\left(\frac{1}{3} - \frac{1}{\sqrt{3}}\right)$ (D) $\left(\frac{1}{4}, -\frac{1}{2}\right)$

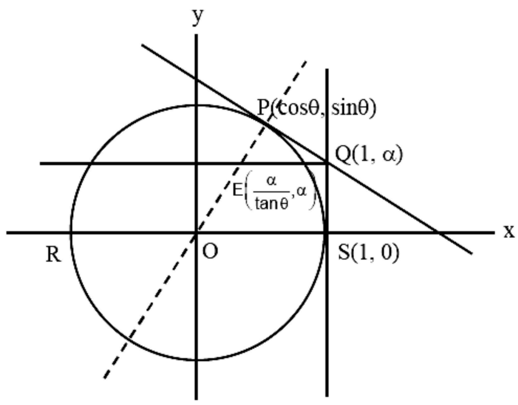
Ans. C, A

Sol. $E = \left(\frac{\alpha}{tan\,\theta}, \alpha\right)$

$\cos\theta + \alpha \sin\theta = 1$

$\alpha = \tan \dfrac{\theta}{2}$

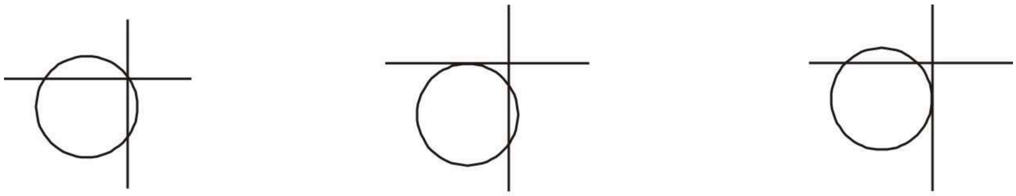
$\therefore$ locus is $y^2 = 1 - 2x$



Q7. For how many values of p, the circle x2 + y2 + 2x + 4y – p = 0 and the coordinate axes have exactly three common points?

Ans. 2

Sol.



p = 0

not possible

2 values of p.

p = –1

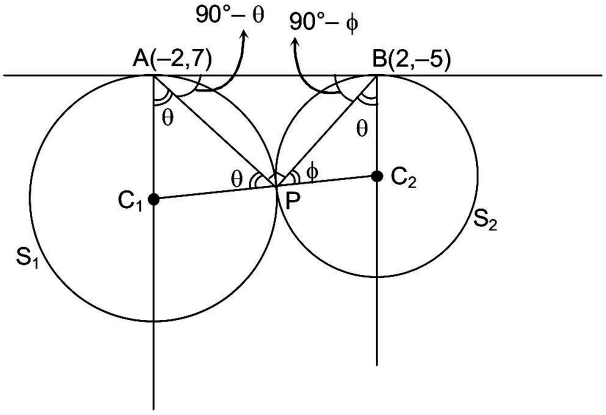
Q8. Let T be the line passing through the points P(–2, 7) and (2, –5). Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangent to S_1 at P, and tangent to S_2 at Q, and also such that S_1 and S_2 touch each other at a point, say, M. Let E1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1. Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point R(1, 1) be F_2. Let E_2 be the set of the mid-points of the line segments in the set F_2. Then, which of the following statement(s) is (are) TRUE?

(A) The point (–2, 7) lies in E_1 (B) The point (4/5, 7/5) does NOT lie in E_2

(C) The point (1/2, 1) lies in E_2 (D) The point (0, 3/2) does NOT lie in E_1

Ans. B, D

Sol.



Note that $\angle APB = \frac{\pi}{2}$, hence locus of P is $(x + 2)(x – 2) + (y – 7)(y + 5) = 0$

$d^2 + y^2 – 2y – 39 = 0$ .... E_1

Locus of mid-points of chords passing through (1, 1) is

$h + K – (1 + k) = h^2 + k^2 – 2K$

$\Rightarrow \quad h^2 + K^2 – 2K – h + 1 = 0$

Hence E_2 is $x^2 + y^2 – x – 2y + 1 = 0$

Paragraph

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$.

(There are two questions based on PARAGRAPH "X", the question given below is one of them)

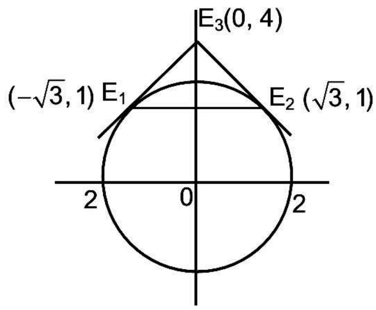
Q9. Let E_1E_2 and F_1F_2 be the chords of S passing through the point P_0 (1,1) and parallel to the x-axis and the y-axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slope –1. Let the tangents to S at E_1 and E_2 meet at E_3, the tangents to S at F_1 and F_2 meet at F_3, and the tangents to S at G_1 and G_2 meet at G_3. Then, the points E_3, F_3, and G_3 lie on the curve :

(A) $x + y = 4$ (B) $(x – 4)2 + (y – 4)2 = 16$ (C) $(x – 4) (y – 4) = 4$ (D) $xy = 4$

Ans. A

Sol. Tangent at E_1 and E_2 are $– \sqrt{3}x + y = 4$ and $\sqrt{3}x + y = 4$

They intersect at E_3 (0, 4)



$F_1(1,)$, F_2 $(1, -\sqrt{3})$, F_3 (4, 0)

$G_1(0, 2)$, $G_2(2, 0)$, $G_3(2, 2)$

E_3 F_3, G_3 lie on line x + y = 4

Paragraph

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$.

(There are two questions based on PARAGRAPH "X", the question given below is one of them)

Q10. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N. Then, the mid-point of the line segment MN must lie on the curve :

(A) $(x + y)^2 = 3$ (B) $x^{2/3} + y^{2/3} = 2^{4/3}$ (C) $x^2 + y^2 = 2xy$ (D) $x^2 + y^2 = x^2 y^2$

Ans. D

Sol. Let P(2 cos q, 2 sin q)

Tangent is x cos q + y sin q = 2

$$M\left(\frac{2}{cos\,\theta}, 0\right), N\left(0, \frac{2}{cos\,\theta}\right)$$

$$x = \frac{1}{cos\,\theta} \text{ and } y = \frac{1}{sin\theta}$$

$$\Rightarrow \quad \frac{1}{x^2} + \frac{1}{y^2} = 1$$

$$\Rightarrow \quad x^2 + y^2 = x^2y^2$$

Q11. A line y = m x + 1 intersects the circle $(x-3)^2 + (y+2)^2 = 25$ at the points P and Q. If the midpoint of the line segment PQ has x–coordinate $-\frac{3}{5}$ then which one of the following options is correct ?

(A) $6 \leq m < 8$ (B) $4 \leq m < 6$ (C) $2 \leq m < 4$ (D) $-3 \leq m < -1$

Ans. C

Sol. Circle $(x - 3)^2 + (y + 2)^2 = 25$

y = mx + 1

Solve

$(x - 3)^2 + (mx + 1 + 2)^2 = 25$

$(1 + m2)x2 + x(6m - 6) - 7 = 0$

$$(1+ m^2)x2 + x(6m- 6)- 7 = 0 \begin{cases} x_1 \\ x_2 \end{cases}$$

$5(m - 1) = m2 + 1$

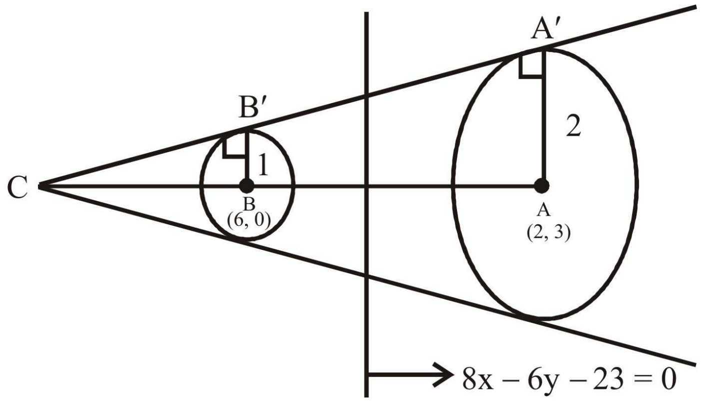
$m2 - 5m + 6 = 0$

$(m - 2)(m - 3) = 0$

Q12. Let the point B be the reflection of the point A(2, 3) with respect to the line $8x - 6y - 23 = 0$. Let Γ_A and Γ_B be circles of radii 2 and 1 with centres A and B respectively. Let T be a common tangent to the circles Γ_A and Γ_B such that both the circles are on the same side of T. If C is the point of intersection of T and the line passing through A and B, then the length of the line segment AC is…..

Ans. 10

Sol. $\frac{x-2}{8} = \frac{y-3}{-6} = -2\left(\frac{16-18-23}{100}\right)\frac{x-2}{8} = \frac{y-3}{-6} = \frac{-2\times-25}{100} = \frac{1}{2}\frac{x-2}{4} = \frac{y-3}{-3} = 1x - 2 = 4 \qquad y -$

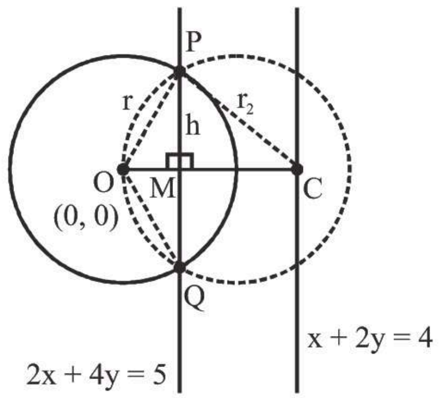


$\frac{CB}{CA} = \frac{1}{2}(\Delta'S\ CBB'and\ CAA'\ are\ \ congruent)\frac{CB}{CB+AB} = \frac{1}{2}2CB = CB + ABCB = AB = \sqrt{16 +}$

Q13. Let O be the centre of the circle $x^2 + y^2 = r^2$, where $r > \frac{\sqrt{5}}{2}$. Suppose PQ is a chord of this circle and the equation of the line passing through P and Q is $2x + 4y = 5$. If the centre of the circumcircle of the triangle OPQ lies on the line $x + 2y = 4$, then the value of r is _____

Ans. 2

Sol.



MC = distance between two lines

$MC = \frac{3}{2\sqrt{5}}$

$OC = r_2 = \frac{4}{5}$

$h^2 + OM^2 = r^2$ -(1)

$h^2 + CM^2 = r_2^2$ -(2)

$h^2 + = \frac{25}{20} = r^2$ -(1)

$h^2 + CM^2 = r_2^2$ -(2)

$h^2 + \frac{9}{20} = \frac{16}{5}$ -(2)

by (1) - (2)

$r^2 = 4$

$r = 2$

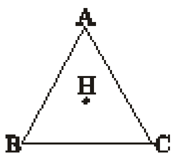
Q14. Consider a triangle Δ whose two sides lie on the x-axis and the line x + y + 1 = 0. If the orthocenter of Δ is (1, 1), then the equation of the circle passing through the vertices of the triangle Δ is

(A) $x^2 + y^2 - 3x + y = 0$ (B) $x^2 + y^2 + x + 3y = 0$ (C) $x^2 + y^2 + 2y - 1 = 0$

(D) $x^2 + y^2 + x + y = 0$

Ans. B

Sol.



equation of AC x + y + 1 = 0

equation of BC y = 0

orthocenter H(1, 1)

equation of AH x = 1

equation of BH $x - y = 0$

Get $A = (1, -2)$; $B(0, 0)$ $C(-1, 0)$

Let circum center $P(h, k)$

$PA = PB$ $PB = PC$

$(h-1)^2 + (k+2)^2 = h^2 + k^2$ $h^2 + k^2 = (h+1)^2 + k^2$

$\Rightarrow h = \frac{-1}{2}; k = -\frac{3}{2}$

equation of circum circle $x^2 + y^2 + x + 3y = 0$

Paragraph

Let $M = \{(x, y) \in R \times R : x^2 + y^2 \le r^2\}$,

\nwhere $r > 0$. Consider the geometric progression $a_n = \frac{1}{2^{n-1}}, n = 1, 2, 3, \ldots$. Let $S_0 = 0$ and, for $n \ge 1$, let S_n denote the sum of the first n terms of this progression. For $n \ge 1$, let C_n denote the circle with center $(S_{n-1}, 0)$ and radius a_n, and D_n denote the circle with center (S_{n-1}, S_{n-1}) and radius a_n.

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Q15. Consider M with $r = \frac{1025}{512}$ .Let k be the number of all those circles Cn that are inside M. Let l be the maximum possible number of circles among these k circles such that no two circles intersect. Then

(A) $k + 2l = 22$ (B) $2k + l = 26$ (C) $2k + 3l = 34$ (D) $3k + 2l = 40$

Ans. D

Sol. $a_n = \frac{1}{2^{n-1}}$

$S_n = 1 + \frac{1}{2} + \ldots\ldots. \frac{1}{2^{n-1}}$

$S_n = 2 - \frac{1}{2^{n-1}}$

hence centre of C_n is $\left(2 - \frac{1}{2^{n-2}}, 0\right)$ and radius of C_n is $\frac{1}{2^{n-1}}$

$r = \frac{1025}{513} < 2$

here C_n inside M then

$d < |r_1 - r_2|$

$2 - \frac{1}{2^{n-2}} < 2 - \frac{1}{2^{n-1}}$

$2 - \frac{1}{2^{n-2}} + \frac{1}{2^{n-1}} < 2$

$\Rightarrow K = 10$

also we observe that $l = 5$ then option (D) is correct.

Paragraph

Let $M = \{(x, y) \in R \times R : x^2 + y^2 \le r^2\}$,

\nwhere $r > 0$. Consider the geometric progression $a_n = \frac{1}{2^{n-1}}, n = 1, 2, 3, \ldots$. Let $S_0 = 0$ and, for $n \geq 1$, let S_n denote the sum of the first n terms of this progression. For $n \geq 1$, let C_n denote the circle with center $(S_{n-1},0)$ and radius a_n, and D_n denote the circle with center (S_{n-1},S_{n-1}) and radius a_n.

\n

Q16. Consider M with $r = \frac{(2^{199}-1)\sqrt{2}}{2^{198}}$. The number of all those circles D_n that are inside M is

(A) 198 (B) 199 (C) 200 (D) 201

Ans. B

Sol. Centre of D_n here $S_n = 2 - \frac{1}{2^{n-1}}$

(S_{n-1}, S_{n-1}) and radius is $\frac{1}{2^{n-1}}$

D_n lies inside M then

$d < |r_1 - r_2|$

$\sqrt{2}\,S_{n-1} < \frac{(2^{199}-1)\sqrt{2}}{2^{198}} - \frac{1}{2^{n-1}}$

$\sqrt{2}\left(2 - \frac{1}{2^{n-2}}\right) < \frac{(2^{199}-1)\sqrt{2}}{2^{198}} - \frac{1}{2^{n-1}}$

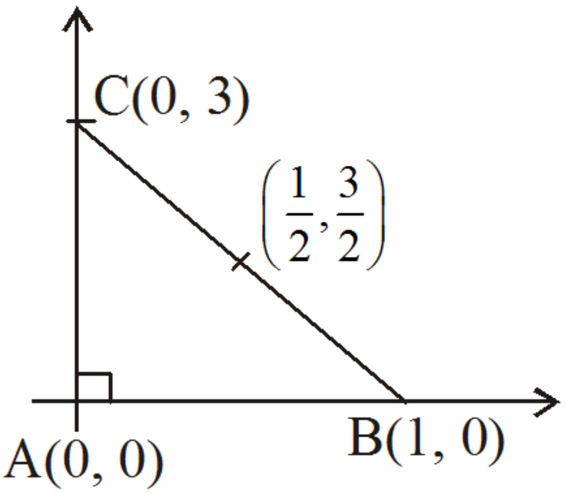
$\frac{\sqrt{2}}{2^{n-2}} > \frac{\sqrt{2}}{2^{198}} + \frac{1}{2^{n-1}}$

$\Rightarrow$ n = 199.

Q17. Let ABC be the triangle with AB = 1, AC = 3 and $\angle BAC = \frac{\pi}{2}$. If a circle of radius r > 0 touches the sides AB, AC and also touches internally the circumcircle of the triangle ABC, then the value of r is ______________.

Ans. 0.83, 0.84

Sol. Let A be the origin B on x-axis, C on y-axis as shown below



$\therefore$ Equation of circumcircle is

$\left(x-\frac{1}{2}\right)^2 + \left(y-\frac{3}{2}\right)^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2 = \frac{5}{2}$ _________(1)

Required circle touches AB and AC, have radius r

$\therefore$ Equation be $(x-r)^2 + (y-r)^2 = r^2$ __________(2)

If circle in equation (2) touches circumcircle internally, we have

$d_{c_1c_2} = |r_1 - r_2|$

$$\Rightarrow \left(\frac{1}{2}-r\right)^2 + \left(\frac{3}{2}-r\right)^2 = \left(\left|\sqrt{\frac{5}{2}}-r\right|\right)^2$$

$$\Rightarrow \frac{1}{4} + r^2 - r + \frac{9}{4} + r^2 - 3r = \left(\sqrt{\frac{5}{2}}-r\right)^2$$

$$\Rightarrow \frac{1}{4} + r^2 - r + \frac{9}{4} + r^2 - 3r = \left(\sqrt{\frac{5}{2}}-r\right)^2 \text{ or } \left(r-\sqrt{\frac{5}{2}}\right)^2$$

$$\Rightarrow 2r^2 - 4r + \frac{5}{2} = \frac{5}{2} + r^2 - \sqrt{10}r$$

$\Rightarrow r = 0$ or $4 - \sqrt{10}$

$\Rightarrow r = 0.837$

Q18. Let G be a circle of radius $R > 0$. Let $G_1, G_2, \ldots, Gn$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \ldots\ldots Gn$ touches the circle G externally. Also, for $i = 1, 2, \ldots, n-1$, the circle Gi touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are TRUE ?

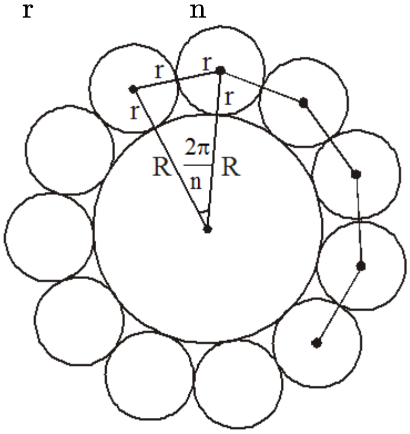
(A) If $n = 4$, then $\left(\sqrt{2}-1\right)r < R$ (B) If $n = 5$, then $r < R$

(C) If $n = 8$, then $\left(\sqrt{2}-1\right)r < R$ (D) If $n = 12$, then $\sqrt{2}\left(\sqrt{3}+1\right)r < R$

Ans. C, D

Sol. $2(R+r)\sin\frac{\pi}{n} = 2r$

$$\frac{R+r}{r} = \text{cosec}\,\frac{\pi}{n}$$



(A) $n = 4, R + r = \sqrt{2}r$

(B) $n = 5, \frac{R+r}{r} = \text{cosec}\,\frac{\pi}{5} < \text{cosec}\,\frac{\pi}{6}$

$R + r < 2r \Rightarrow r > R$

(C) $n = 8, \frac{R+r}{r} = \text{cosec}\,\frac{\pi}{8} > \text{cosec}\,\frac{\pi}{4}$

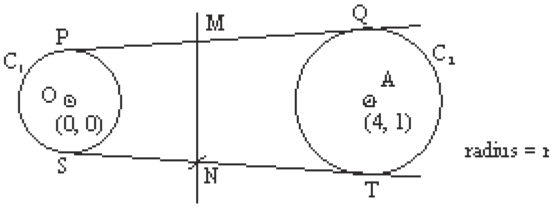
(D) $n = 12, \frac{R+r}{r} = \text{cosec}\,\frac{\pi}{12} = \sqrt{2}(\sqrt{3}+1)$

$$R + r = \sqrt{2}(\sqrt{3}+1)r$$
$$\sqrt{2}(\sqrt{3}+1)r > R$$

Q19. Let C_1 be the circle of radius 1 with center at the origin. Let C_1 be the circle of radius r with center at the point A = (4, 1), where 1 < r < 3. Two distinct common tangents PQ and ST of C_1 and C_2 are drawn. The tangent PQ touches C_1 at P and C_2 at Q. The tangent ST touches C_1 at S and C_2 at T. Mid points of the line segments PQ and ST are joined to form a line which meets the x–axis at a point B. If $AB = \sqrt{5}$ then the value of r^2 is

Ans. 2

Sol.



Let the tangents are DCT's

line MN will be radical Axis as PM = PQ and SN = NT

$C_1 : x^2 + y^2 = 1$

$C_2 : (x - 4)^2 + (y - 1)^2 = r^2$

$RA : S_1 - S_2 = 0$

$\Rightarrow 8x + 2y = 18 - r^2$

Point B is on x-axis

put y = 0 $\quad \Rightarrow x = \dfrac{18 - r^2}{8}$

So co-ordinates of $B\left(\dfrac{18-r^2}{8}, 0\right)$

Given AB = $\sqrt{5}$ $\quad \Rightarrow AB^2 = 5$

$\Rightarrow \left(\dfrac{18-r^2}{8} - 4\right)^2 + 1 = 5$

$\Rightarrow \quad \dfrac{18 - r^2 - 32}{8} = \pm 2$

$\Rightarrow \; r^2 + 14 = \pm 16$

$r^2 = -30$ (rejected)

$r^2 = 2$

Ans. 2

Q20. Let the straight line y = 2x touch a circle with center (0, α), α > 0, and radius r at a point A_1. Let B_1 be the point on the circle such that the line segment A_1B_1 is a diameter of the circle. Let $\alpha + r = 5 + \sqrt{5}$.

Match each entry in List-I to the correct entry in List-II.

List-I	List-I

(P) α equals	(1) (–2, 4)
(Q) r equals	(2) $\sqrt{5}$
(R) A_1 equals	(3) (–2, 6)
(S) B_1 equals	(4) 5
	(5) (2,4)

The correct option is :

(A) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

(C) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (3)

(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)

Ans. 66618eea601efe92aa062625

Sol. Circle $(x - 0)^2 + (y - a)^2 = r^2$

tangent $2x - y = 0$

by condition of tangency

$$\left|\frac{0-\alpha}{\sqrt{5}}\right| = r$$

$|-\alpha| = r\sqrt{5}$

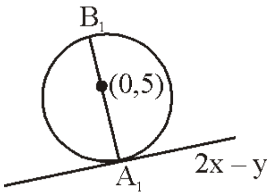
$\alpha = \pm r\sqrt{5}$

$\alpha = r\sqrt{5}$ (a > 0) ___________(1)

given $\alpha + r = 5 + \sqrt{5}$ ___________(2)

by (1) and (2)

$\alpha = 5$, $r = \sqrt{5}$



Circle $x^2 + (y - 5)^2 = 5$

tangent $2x - y = 0$

then A_1 (2, 4)

Clearly B_1(–2, 6)

Hence, correct option is (C).

Exercise C1

Q1. If a > 2b > 0 then the positive value of m for which $y = mx - b\sqrt{1+m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x - a)^2 + y^2 = b^2$ is

(A) $\frac{2b}{\sqrt{a^2-4b^2}}$ (B) $\frac{\sqrt{a^2-4b^2}}{2b}$ (C) $\frac{2b}{a-2b}$ (D) $\frac{b}{a-2b}$

Ans. A

Sol. C.O.T

$$(x-a)^2 = y^2 = b^2$$

$$mx - y - b\sqrt{1+m^2} = 0$$

$$\left|\frac{am - b\sqrt{1+m^2}}{\sqrt{1+m^2}}\right| = b$$

$$a^2m^2 + b^2(1+m^2) - 2amb\sqrt{1+m^2} = b^2(1+m^2)$$

$$a^2m^2 = 2abm\sqrt{1+m^2}$$

$$m = 0 \qquad \text{or} \qquad a^2m^2 = 4ab^2(1+m^2)$$

$$a^2m^2 - 4b^2m^2 = 4b^2$$

$$m^2 = \frac{4b^2}{a^2-4b^2}$$

$$m \pm \frac{2}{\sqrt{a^2-4b^2}}$$

Q2. A circle touches a straight *l*ine lx + my + n = 0 & cuts the circle $x^2 + y^2 = 9$ orthogonally. The locus of centres of such circles is:

(A) $(lx + my + n)^2 = (l^2 + m^2)(x^2 + y^2 - 9)$ (B) $(lx + my - n)^2 = (l^2 + m^2)(x^2 + y^2 - 9)$

(C) $(lx + my + n)^2 = (l^2 + m^2)(x^2 + y^2 + 9)$ (D) none of these

Ans. A

Sol. Let required equation of circle is x2 + y2 + 2gx + 2gx + 2fy + c = 0
it cuts the circle x2 + y2 – 9 = 0 orthogonally
$\therefore$ 2g(0) + 2f(0) = c – 9 $\Rightarrow$ c = 9
It also touches straight line lx + my + n = 0

$$\therefore \left|\frac{l(-g) + m(-f) + n}{\sqrt{l^2+m^2}}\right| = \sqrt{g^2+f^2-9}$$

Locus of centre (–g, –f) is $(lx + my + n)^2 = (x^2 + y^2 - 9)(l^2 + m^2)$

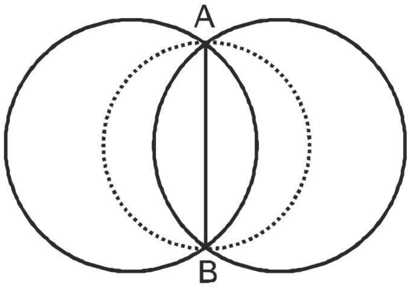
Q3. The circle $x^2 + y^2 = 4$ cuts the circle $x^2 + y^2 + 2x + 3y - 5 = 0$ in A & B. Then the equation of the circle on AB as a diameter is:

(A) $13(x^2 + y^2) - 4x - 6y - 50 = 0$ (B) $9(x^2 + y^2) + 8x - 4y + 25 = 0$

(C) $x^2 + y^2 - 5x + 2y + 72 = 0$ (D) none of these

Ans. A

Sol.



Common chord of given circle

$2x + 3y - 1 = 0$

family of circle passing through point of intersection of given circle

$(x^2 + y^2 + 2x + 3y - 5) + \lambda(x^2 + y^2 - 4) = 0 \Rightarrow (\lambda + 1)x^2 + (\lambda + 1)y^2 + 2x + 3y - (4\lambda + 5) = 0$

$$x^2 + y^2 + \frac{2x}{\lambda+1} + \frac{3}{\lambda+1}y - \frac{(4\lambda+5)}{\lambda+1} = 0$$

centre $\left(-\frac{1}{\lambda+1}, \frac{-3}{2(\lambda+1)}\right)$

This centre lies on AB

$$2\left(-\frac{1}{\lambda+1}\right) + 3\left(\frac{-3}{2(\lambda+1)}\right) - 1 = 0 \Rightarrow -4 - 9 - 2\lambda - 2 = 0 \Rightarrow \lambda = -\frac{15}{2}$$

$$\left(-\frac{15}{2} + 1\right)x^2 + \left(-\frac{12}{2} + 1\right)y^2 + 2x + 3y - \left(4\times\frac{15}{2} + \right) = 0 \Rightarrow -\frac{13x^2}{2} - \frac{13y^2}{2}$$

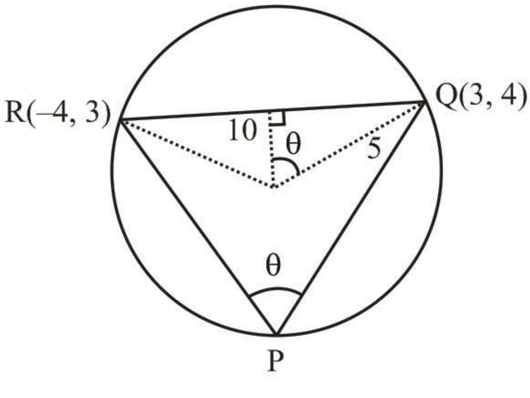
$$\Rightarrow 13(x^2 + y^2) - 4x - 6y - 50 = 0$$

Q4. The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have co–ordinates (3, 4) and (–4, 3) respectively, the Ð QPR is equal to

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

Ans. C

Sol.



equation of QR

$$y-4=\frac{1}{7}(x-3)$$

$$7y-28=x-3$$

$$x-7y+25-0$$

$$p=\left|\frac{25}{\sqrt{50}}\right|=\frac{5}{\sqrt{2}}$$

$$cos\theta=\frac{\frac{5}{=\sqrt{2}}}{5}=\frac{1}{\sqrt{2}}$$

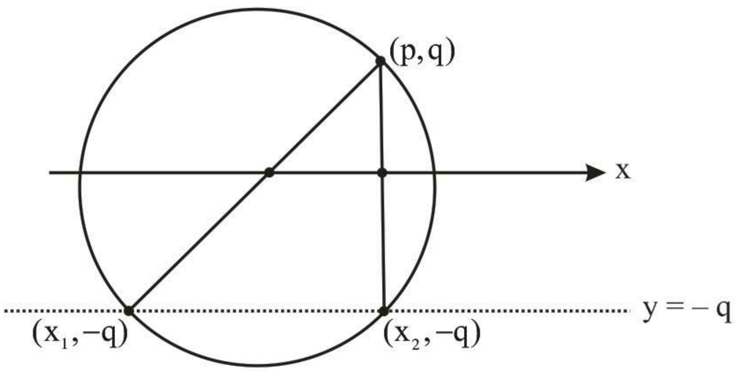
$$\theta=45^{\circ}$$

Q5. If two distinct chords, drawn from the point (p, q) on the circle $x^2 + y^2 = px + qy$ (where $pq \neq 0$) are bisected by the x–axis, then

(A) $p^2 = q^2$ (B) $p^2 = 8q^2$ (C) $p^2 < 8q^2$ (D) $p^2 > 8q^2$

Ans. D

Sol.



to get x_1 and x_2 put $y = -q$ in $x^2 + y^2 = px + qy$

$$x^2+q^2=px-q^2$$

$$x^2-px+2q^2=0$$

x_1 are x_2 are distinct

$D > 0$

$$P^2>8q^2$$

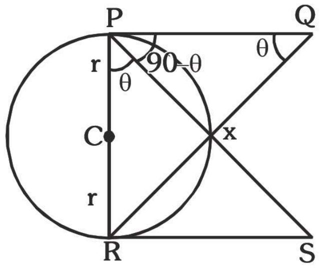
Q6. Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle then 2r equals

(A) $\sqrt{PQ.RS}$ (B) $\frac{PQ+RS}{2}$ (C) $\frac{2PQ+RS}{PQ+RS}$ (D) $\sqrt{\frac{(PQ)^2+(RS)^2}{2}}$

Ans. A

Sol. Let $\angle RPS=\theta$

$\angle XPQ = 90 - \theta$



$\angle RPS = \theta (\because \angle PXQ = 90)$

$\therefore \triangle PRS \sim \therefore \triangle QPR$ (AAA similarity)

$\therefore \dfrac{PR}{QP} = \dfrac{RS}{PR} \Rightarrow PR^2 = PQ.RS$

$\Rightarrow PR = \sqrt{PQRS}$

Q7. A circle is given by $x^2 + (y-1)^2 = 1$, another circle C touches it externally and also the x–axis, then the locus of its centre is

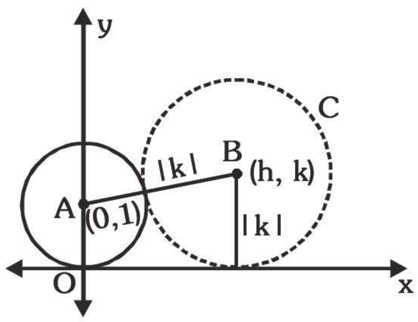
(A) $\{(x, y) : x^2 = 4y\} \cup \{(x, y) : y \leq 0\}$

(B) $\{(x, y) : x^2 + (y-1)^2 = 4\} \cup \{x, y) : y \leq 0\}$

(C) $\{(x, y) : x^2 = y\} \cup \{(0, y) : y \leq 0\}$ (D) $\{(x, y) : x^2 = 4y\} \cup \{(0, y) : y \leq 0\}$

Ans. D

Sol. Let the centre of circle C be (h, k). Then as this circle touches axis of x its radius = |k|



Also it touches the given circle $x^2 + (y-1)^2 = 1$, centre (0, 1) radius 1, externally

Therefore

The distance between centres = sum of radii

$\Rightarrow \sqrt{(h-0)^2 + (k-1)^2} = 1 + |k|$

$\Rightarrow h^2 + k^2 - 2k + 1 = (1 + 1|k|)^2$

$\Rightarrow h^2 + k^2 - 2k + 1 = 1 + 2|k| + k^2$

$\Rightarrow h^2 = 2k + 2|k|$

$\therefore$ Locus of (h, k) is, $x^2 = 2y + 2|y|$

Now if y > 0 it becomes $x^2 = 4y$ and if $y \leq 0$, it becomes x = 0

$\therefore$ Combining the two, the required locus is
$\{(x, y) : x^2 = 4y\} \cup \{(0, y) : y \leq 0\}$

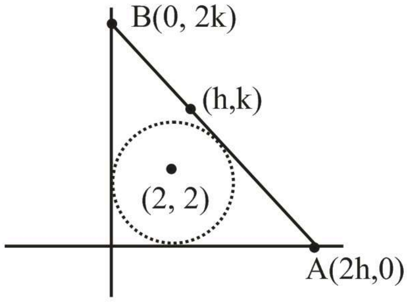
Q8. A circle with center (2,2) touches the coordinate axes and a straight line AB where A and B lie on positive direction of coordinate axes such that the circle lies between origin and the line AB. If O be the origin then the locus of circumcenter of ΔOAB will be :

(A) $xy = x + y + \sqrt{x^2 + y^2}$ (B) $xy = x + y - \sqrt{x^2 + y^2}$ (C) $xy + x + y = \sqrt{x^2 + y^2}$

(D) $xy + x + y + \sqrt{x^2 + y^2} = 0$

Ans. A

Sol.



$$\frac{x}{2h} + \frac{y}{2k} = 1$$

Apply C.O.T.

$$\left|\frac{\frac{1}{h} + \frac{1}{k} - 1}{\sqrt{\frac{1}{4h^2} + \frac{1}{4k^2}}}\right| = 2$$

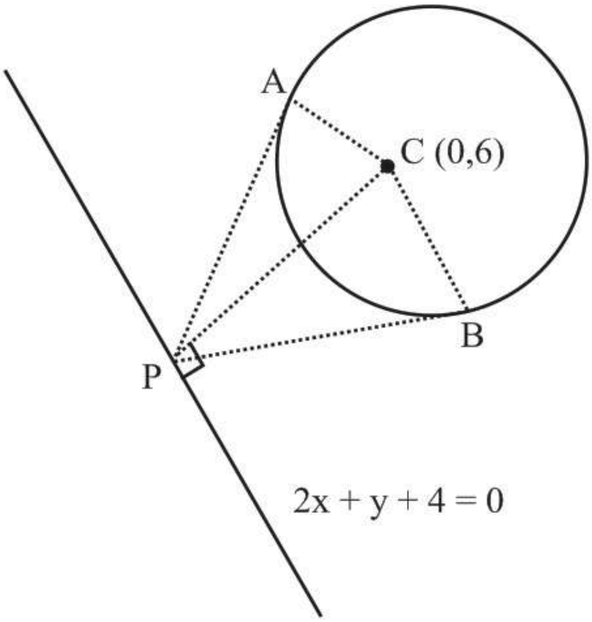
$$\left(\frac{1}{h} + \frac{1}{k} - 1\right)^2 = \frac{1}{h^2} + \frac{1}{k^2}$$

Q9. From a point 'P' on the line 2x + y + 4 = 0; which is nearest to the circle $x^2 + y^2 - 12y + 35 = 0$, tangents are drawn to given circle. The area of quadrilateral PACB (where 'C' is the center of circle and PA & PB are the tangents) is :

(A) 8 (B) $\sqrt{110}$ (C) $\sqrt{19}$ (D) None of these

Ans. C

Sol.



equation of OP

$y-6=\frac{1}{2}(x-0)$

$2y-12=x$

$x-2y+12=0$

$P(-4,4)$

$PA=\sqrt{S}=\sqrt{19}$

Area of $\triangle PAC=\frac{1}{2}\times\sqrt{19}$

Area of $\neq$ PABC = $\sqrt{19}$

Q10. A square OABC is formed by line pairs xy = 0 and xy + 1 = x + y where 'O' is the origin. A circle with center C_1 inside the square is drawn to touch the line pair xy = 0 and another circle with centre C_2 and radius twice that of C_1, is drawn to touch the circle C_1 and the other line pair. The radius of the circle with centre C_1 is :

(A) $\frac{\sqrt{2}}{\sqrt{3}(\sqrt{2}+1)}$ (B) $\frac{2\sqrt{2}}{3(\sqrt{2}+1)}$ (C) $\frac{\sqrt{2}}{3(\sqrt{2}+1)}$ (D) $\frac{\sqrt{2}+1}{3\sqrt{2}}$

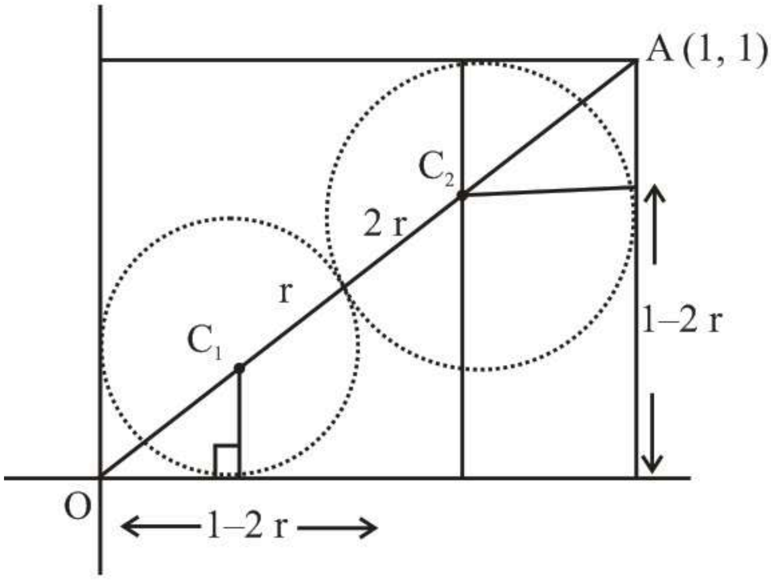
Ans. C

Sol. $xy-x-y+1=0$

$x(y-1)-1(y-1)=0$

$(x-1)(y-1)=0$ and $xy=0$

$x=1, y=1, x=0, y=0$



$C_2(1-2r, 1-2r), \quad C_1(r, r)$

$OA = \sqrt{2}$

$\sqrt{2}r + 3r + 2\sqrt{2}r = \sqrt{2}$

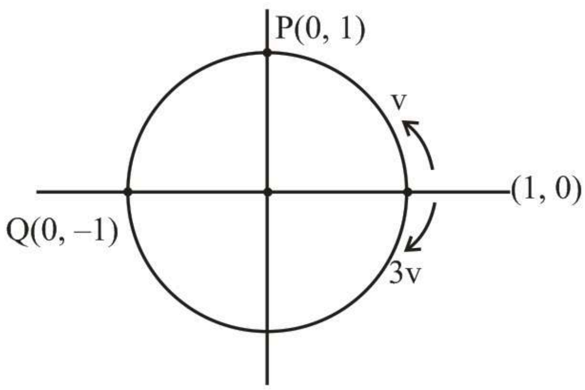
$r = \dfrac{\sqrt{2}}{3(\sqrt{2}+1)}$

Q11. A circle of radius unity is centred at origin. Two particles start moving at the same time from the point (1,0) and move around the circle in opposite direction. One of the particle moves counter clockwise with constant speed v and the other moves clockwise with constant speed 3v. After leaving (1,0), the two particles meet first at a point P, and continue until they meet next at point Q. The coordinates of the point Q are :

(A) (1,0) (B) (0,1) (C) (0,–1) (D) (–1,0)

Ans. D

Sol.

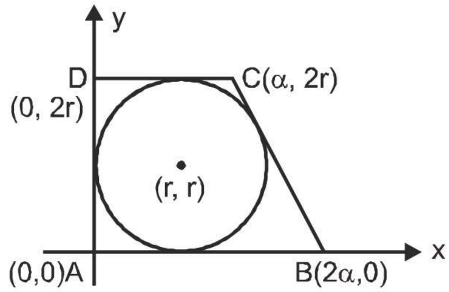


Q12. Let ABCD be a quadrilateral with area 18, with side AB parallel to the side CD and AB = 2CD. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral ABCD touching all the sides, then its radius is

(A) 3 (B) 2 (C) 3/2 (D) 1

Ans. B

Sol.



$18 = \frac{1}{2}(3\alpha)(2r) \Rightarrow \alpha r = 6$

Line, $y = \frac{2r}{\alpha}(x - 2\alpha)$ is tangent to circle

$(x-r)^2 + (y-r)^2 = r^2$

$2\alpha = 3r$ and $\alpha r = 6$

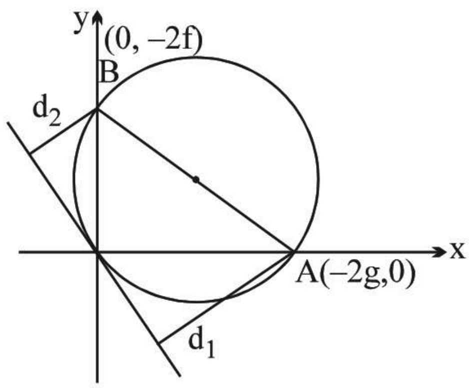
$r = 2$

Q13. A line meets the co-ordinate axes in A and B. A circle is circumscribed about the triangle OAB. If d1 and d_2 are the distances of the tangent to the circle at the origin O from the points A and B respectively, the diameter of the circle is :

(A) $\frac{2d_1 + d_2}{2}$ (B) $\frac{d_1 + 2d_2}{2}$ (C) $d_1 + d_2$ (D) $\frac{d_1 d_2}{d_1 + d_2}$

Ans. C

Sol.



Let the circle be $x^2 + y^2 + 2gx + 2fy = 0$

Tangent at the origin is $gx + fy = 0$

$d_1 = \frac{2g^2}{\sqrt{g^2 + f^2}}$

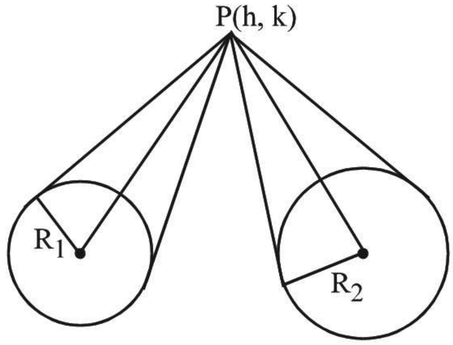
$$d_2 = \frac{2g^2}{\sqrt{g^2+f^2}} \quad \Bigg\} \Rightarrow d_1 + d_2 = 2\sqrt{g^2+f^2} = \text{diameter}$$

Q14. The locus of the point from which two given unequal circle subtend equal angles is :

(A) a straight line (B) a circle (C) a parabola (D) none

Ans. B

Sol.



Use $\frac{R_1}{L_1} = \frac{R_2}{L_2}$

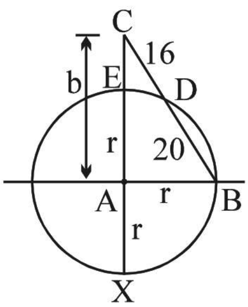
$R_1 L_2 = R_2 L_1$

Q15. Triangle ABC is right angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E respectively. If BD = 20 and DC = 16 then the length AC equals

(A) $6\sqrt{21}$ (B) $6\sqrt{26}$ (C) 30 (D) 32

Ans. B

Sol.



$b^2 + r^2 = (36)^2$ ....(1)

Also CD · CB = CE · CX (using power of the point C)

$16 \cdot 36 = (b - r)(b + r)$

$\therefore b^2 - r^2 = 16 \cdot 36$

from (1) and (2)

$2b^2 = 36(36 + 16) = 36 \cdot 52$

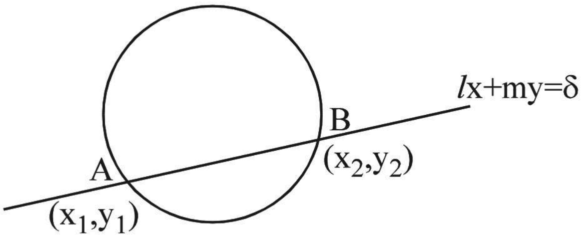
$b^2 = 36 \cdot 26 \Rightarrow b = 6\sqrt{26}$ **Ans.**

Q16. If H represent the harmonic mean between the abscissae, and K that between the ordinates of the points, in which a circle $x^2 + y^2 = c^2$ is cut by a chord $lx + my = d$, where l and m are the direction cosines of the unit vector in the xy plane, then $lH + mK$ has the value equal to (Take: $l^2 + m^2 = 1$)

(A) $2\delta - \frac{c^2}{\delta}$ (B) $\delta - \frac{c^2}{2\delta}$ (C) $\delta - \frac{2c^2}{\delta}$ (D) $2\delta - \frac{c^2}{2\delta}$

Ans. A

Sol.



Solving line and circle

$m^2x^2 + (\delta - lx)^2 = m^2c^2$

$(m^2 + l^2)x2 - 2ldx + (\delta^2 - m^2c^2) = 0 \quad (l2 + m2 = 1) <^{x_1}_{x_2} \; (l^2 + m^2 = 1)$

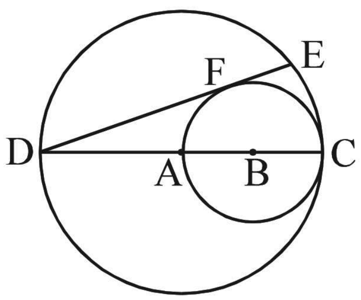
given $H = \frac{2x_1x_2}{x_1 + x_2} = \frac{2(\delta^2 - m^2c^2)}{\frac{2}{\delta}}$

$\Rightarrow lH = \frac{\delta^2 - m^2c^2}{\delta}$ ......(1)

|||ly $mK = \frac{\delta^2 - c^2l^2}{\delta}$ ......(2)

(1) + (2) $\quad lH + mK = \frac{2\delta^2 - c^2(l^2 + m^2)}{\delta} = 2\delta - \frac{c^2}{\delta}$ where $(l^2 + m^2 = 1)$ **Ans.**

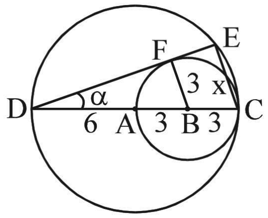
Q17. In the diagram, DC is a diameter of the large circle centered at A, and AC is a diameter of the smaller circle centered at B. If DE is tangent to the smaller circle at F and DC = 12 then the length of DE is



(A) $8\sqrt{2}$ (B) 16 (C) $9\sqrt{2}$ (D) $10\sqrt{2}$

Ans. A

Sol.



$sin\alpha = \frac{3}{9}$

also $sin\alpha = \frac{x}{12}$ (where EC = x)

$\frac{1}{3} = \frac{x}{12} \Rightarrow$ x = 4

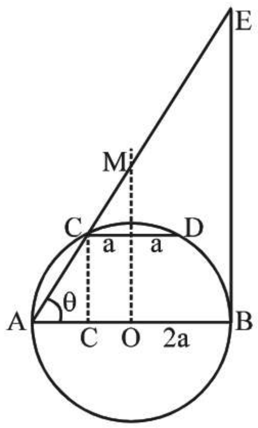
$(DE)^2 = 144 - 16 = 128 \Rightarrow DE = 8\sqrt{2}$

Q18. AB is a diameter of a circle. CD is a chord parallel to AB and 2 CD = AB. The tangent at B meets the line AC produced at E then AE is equal to :

(A) AB (B) $\sqrt{2}$ AB (C) $2\sqrt{2}$ AB (D) 2 AB

Ans. D

Sol.



$x^2 + y^2 = 4a^2$

$(-a, a\sqrt{3}); \tan\theta = \frac{a\sqrt{3}}{a} = \sqrt{3} \Rightarrow \theta = 60^o$

$cos60^o = \frac{4a}{AE}$

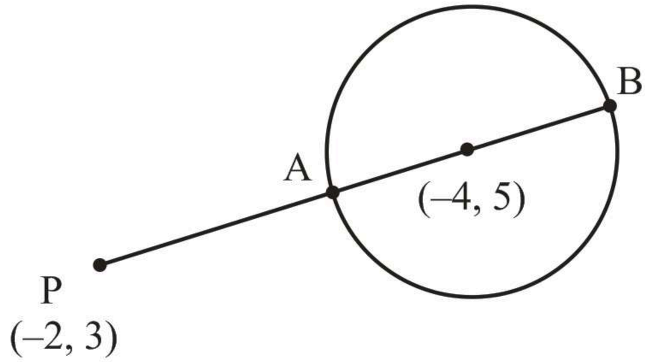
AE = 8a = 2AB $\Rightarrow$ D

Q19. If a = max $\{(x + 2)^2 + (y - 3)^2\}$ and b = min $\{(x + 2)^2 + (y - 3)2\}$ where x,y satisfying $x^2 + y^2 + 8x - 10y - 40 = 0$, then :

(A) a + b = 18 (B) a + b = 178 (C) $a - b = 4\sqrt{2}$ (D) $a - b = 72\sqrt{2}$

Ans. B, D

Sol.



$a = PB^2$ and $b = PA^2$

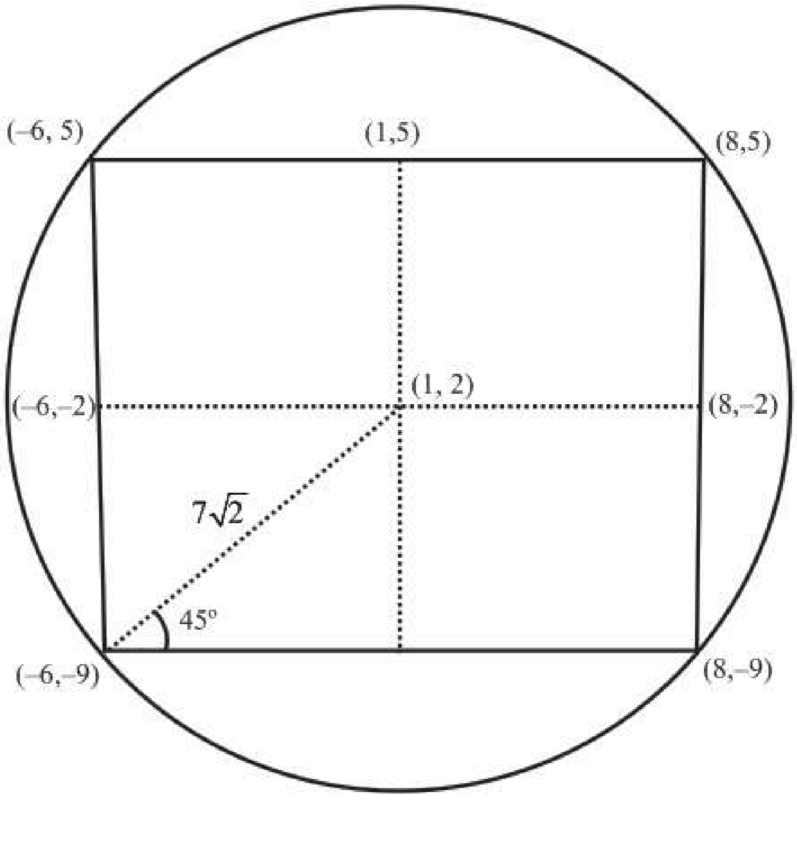
$a = (9 + 2\sqrt{2})^2$ and $b = (9 - 2\sqrt{2})^2$

Q20. A square is inscribed in the circle $x^2 + y^2 - 2x + 4y - 93 = 0$ with the sides parallel to the co–ordinate axes. The co–ordinate of the vertices are :

(A) (8, 5) (B) (8, 9) (C) (–6, 5) (D) (–6, –9)

Ans. A, C

Sol.

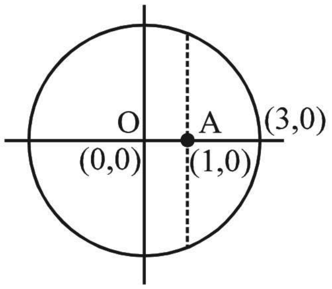


Q21. The centre(s) of the circle(s) passing through the points (0, 0) , (1, 0) and touching the circle $x^2 + y^2=9$ is/are :

(A) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, \frac{3}{2}\right)$ (C) $\left(\frac{1}{2}, 2^{1/2}\right)$ (D) $\left(\frac{1}{2}, -2^{1/2}\right)$

Ans. D, C

Sol.



consider family of circles through (0, 0) and (1, 0)

$x(x-1)+y^2+\lambda y=0$

touches $x^2+y^2=9$

$\therefore$ common chord $= -x+hy+9=0$ ....(1)

$\therefore$ perpendicular from (0, 0) on (1) is equal to 3.

$\left|\frac{9}{\sqrt{1+\lambda^2}}\right|=3 \Rightarrow \lambda^2=\pm 2\sqrt{2}$

circle $x(x-1)+y^2+2\sqrt{2}\,y=0$

$\therefore$ centre $\left(\frac{1}{2},-2^{-\frac{1}{2}}\right)$

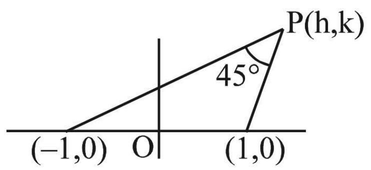
Q22. Locus of the intersection of the two straight lines passing through (1, 0) and (–1, 0) respectively and including an angle of 45° can be a circle with

(A) centre (1, 0) and radius $\sqrt{2}$. (B) centre (1, 0) and radius 2.

(C) centre (0, 1) and radius $\sqrt{2}$. (D) centre (0, – 1) and radius $\sqrt{2}$.

Ans. D, C

Sol.



$m_1=\frac{k}{h-1}; m_2=\frac{k}{h+1}$

$\therefore tan45^{\circ}=\left|\frac{\frac{k}{h-1}-\frac{k}{h+1}}{1+\frac{k^2}{h^2-1}}\right|=\left|\frac{k(h+2-h+1)}{h^2-1+k^2}\right|$

$h^2+k^2-1=\pm 2k$

locus is $x^2+(y\pm 1)^2=(\sqrt{2})^2$

$(0,1)$ or $(0,-1)$ and radius $=\sqrt{2}$

$\Rightarrow$ (C) and (D) Ans.

Q23. The circles $x^2+y^2+2x+4y-20=0$ & $x^2+y^2+6x-8y+10=0$

(A) are such that the number of common tangents on them is 2 (B) are orthogonal

(C) are such that the length of their common tangent is $5\,(12/5)^{1/3}$

(D) are such that the length of their common chord is $5\sqrt{\frac{3}{2}}$

Ans. A, B, D

Sol. (A) $C_1 : (-1, -2), r_1 = 5$

$C_2 : (-3, 4), r_2 = \sqrt{15}$

$|r_1 - r_2| < C_1C_2 = \sqrt{40} < r_1 + r_2$

(B) For orthogonality

$2(1 \times 3 + 2\,(-4)) = -10$

and $c_1 + c_2 = -10$

(C) $L_{CT} = \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2}$

(D) Equation of common chord : $2x - 6y + 15 = 0$

Q24. Consider the circle $x^2 + y^2 - 8x - 18y + 93 = 0$ with centre 'C' and point $P(2, 5)$ outside it.

From the point P, a pair of tangents PQ and PR are drawn to the circle with S as the midpoint of QR. The line joining P to C intersects the given circle at A and B. Which of the following hold(s) good?

(A) CP is the arithmetic mean of AP and BP.

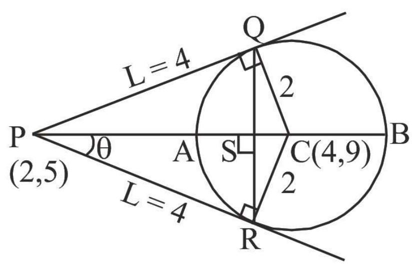
(B) PR is the geometric mean of PS and PC.

(C) PS is the harmonic mean of PA and PB.

(D) The angle between the two tangents from P is $\tan^{-1}\left(\frac{3}{4}\right)$.

Ans. A, C, B

Sol.



Radis = $\sqrt{16 + 81 - 93} = 2$

$CP = \sqrt{20}$; $AP = \sqrt{20} - 2$; $BP = \sqrt{20} + 2$

$\Rightarrow CP = \frac{AP + BP}{2} \Rightarrow$ (A) is correct

Now, let L = PR = $\sqrt{(\mathrm{FC})^2 - \mathrm{r}^2} = \sqrt{20-4} = 4 = \mathrm{PQ}$; $tan\theta = \frac{2}{4} = \frac{1}{2}$

Also $cos\theta = \frac{PS}{PR} \quad \Rightarrow \mathrm{PS} = \mathrm{PR}\cos\theta = 4.\left(\frac{2}{\sqrt{5}}\right) = \frac{8}{\sqrt{5}}$

Harmonic Mean between PA and PB = $\frac{2(\sqrt{20}-2)\sqrt{20}+2)}{2\sqrt{20}} = \frac{16}{2\sqrt{5}} = \frac{8}{\sqrt{5}} = PS \Rightarrow$ (C) is correct

Hence (PS)(PC) = $\left(\frac{8}{\sqrt{5}}\right)(\sqrt{20}) = 16 = (PR)^2$

$\Rightarrow$ PR is the Geometric Mean of PS and PC $\Rightarrow$ (B) is correct

Now angle between the two tangents is 2θ, then (As $tan\theta = \frac{1}{2}$)

$2tan^{-1}\left(\frac{1}{2}\right) = tan^{-1}\left(\frac{2.\frac{1}{2}}{1-\frac{1}{4}}\right) = tan^{-1}\left(\frac{4}{3}\right) \Rightarrow$ (D) is correct